

Ziqi Cao

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EXPERIENCE

Carlson Analytics Lab

Data Analytics Consultant

Minneapolis, USA

Aug 2025 – Aug 2026

Client: Medtronic, Cardiac Intelligence Exchange (CIX)

- Cut refresh time by **3+ hours** by building a Python **ETL pipeline** to extract 20,000+ ClinicalTrials.gov records, standardizing sponsor names via rule-based mapping and tracking changes with row-hash diffing across 12+ relational tables.
- Surfaced **\$2M+ in competitive intelligence value**, previously requiring **3 months** of manual review, by designing a two-component **weighted scoring model** combining rule-based **feature engineering** in Medtronic GPT to track 6 key competitors' activities and potential future market signals.
- Cut **reporting costs by 30%** and ensured **zero unsupported scores** by performing **sensitivity analysis** and **robustness testing** across trial conditions, building an LLM guardrail layer to validate outputs.

Client: A Leading Construction Firm

- Identified **\$4M in potential revenue** upside and **264 recoverable analyst hours** by ingesting and cleaning 3,500+ public bid records using **KNN imputation**, and building predictive models to support bid strategy recommendations.
- Achieved an **80% match rate** against the firm's existing key competitor list, isolating 28 competitors with systematic underbidding or overbidding patterns, by applying **LOF outlier detection** to identify anomalous competitor bids.
- Reduced bid **cost exposure by 90 bps** by running **logistic regression** to quantify competitor-specific win probabilities, translating significant project-scale and geographic pricing effects into go/no-go recommendations.

Client: A K-12 Transportation Management Firm

- Cut reporting effort by **2 days** by building scalable automated daily monitoring on 2.35M+ real-time GPS positions across 14,800 trips to validate route execution and flag missed stops, and standardizing log formats for consistency.
- Reduced **data quality risks by 12%** by applying **Principal Component Analysis** to consolidate vendor coverage quality metrics into composite risk scores, ranking 5 vendors by operational performance and enabling targeted monitoring.

AI Agent: RAG-Based Clinical Note Intelligence System

- Reduced query time by **99%** by developing an end-to-end **RAG pipeline** over 40,000+ unstructured MIMIC-IV discharge notes using **Python**, FAISS vector search, and section-level metadata, enabling semantic search across patient cohorts.
- Achieved **100% source traceability** while **minimizing hallucinations** by building a GPT-4-powered **agentic retrieval system** in Streamlit, routing user queries to source-cited answers with referenced note IDs and confidence scores.

Development Bank of Singapore

Data Analyst, Private Banking

Shanghai, China

Jul 2024 – Jul 2025

- Generated **\$2.16M in QDII sales** and **\$43K in profit within two months** by building **Ridge regression** models on macroeconomic indicators to forecast short-term portfolio performance and informing client recommendations.
- Contributed to **32% monthly AUM growth** by developing **statistical models** (CAPM) to analyze 100+ client portfolios, isolating alpha from market returns to identify outperforming holdings and **support allocation recommendations**.
- Cut manual **reporting time by 10 hours** by leveraging **SQL** to consolidate fund performance data across 10+ fund managers into unified **Tableau dashboards**, enabling client-facing portfolio reviews.
- Conducted A/B testing on client-facing app in partnership with Product and IT teams, optimizing button placement and product information display, reducing drop-off rate from 75% to 63% and increasing product click-through rate by 15%.

Guotai Junan Securities Co., Ltd.

Data Analyst, Institutional Client Dept.

Shanghai, China

Jul 2023 – Jan 2024

- Enabled recurring reporting for **\$720K+ in transaction volume** by **standardizing** onboarding data for 4 wealth management institutions, mapping account fields, fee thresholds, and transaction eligibility rules into internal database.
- Reduced **processing errors to 2%** and **saved 3 reporting hours** weekly by analyzing \$50K+ in failed transaction authorizations and automating reconciliation using **Power Query**, **XLOOKUP**, and **VBA**.

EDUCATION

Carlson School of Management, University of Minnesota, MN

Aug 2025 – Aug 2026

- Master of Science in Business Analytics | **GPA: 3.95/4.00** | **Rank: 1/40**

Renmin University of China, Beijing, China

Sep 2022 – Jun 2024

- Master of Management in Public Administration

Tongji University, Shanghai, China

Sep 2017 – Jun 2022

- Bachelor of Economics in International Trade (Transferred from English Major) | **GPA: 4.71/5.00** | **Rank: 1/25**

SKILLS

- **Business Strength:** CFA Level I, FRM Level I&II (Certified)
- **Strategic & Quantitative Analysis:** A/B Test, Hypothesis Testing, Exploratory Data Analysis, Predictive Modeling, Machine Learning, Statistical Modeling, Time Series Analysis, Root Cause Analysis, Causal Inference, Visualization
- **Data & Visualization Tools:** SQL, Python, R, Tableau, Power BI, Advanced Excel, Databricks, AWS
- **Communication & Leadership:** Debate Skills (2× school debate champion)